PYTHON CAMPUS CHALLENGE

PROBLEM SET 1 • CASE-BASED QUESTIONS

Python Session Series

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SESSIONS COVERED	Python Without Fear + Making Python Think
TOTAL QUESTIONS	16

INSTRUCTIONS

Attempt all questions. Write Python code using only the concepts covered in the two sessions: `print`, comments, variables, data types, numbers, float, string, boolean, arithmetic, `input`, comparison operators, logical operators, `if-elif-else`, `for` loop, `range`, lists, and filtering with `for + if`.

CHALLENGE MODE
Read each case carefully, identify the condition, and write clear, readable Python code.

PROBLEM SET 1 — QUESTIONS

Use the concepts from the sessions. Keep your code simple and add comments wherever asked.

Q1. Smart Q Discount Bill

You are ordering food from the Smart Q app in the cafeteria. Ask for the student's name, order amount, and loyalty points.
• If amount is greater than 200 **and** points are greater than 50, give 20% discount.
• Else if amount is greater than 100, give 10% discount.
• Else give no discount.
Print the final amount with proper comments.

Q2. Affordable Cafeteria Menu

The cafeteria menu is `["Samosa", "Poha", "Maggi", "Tea", "Cold Coffee"]`.
Prices are `[20, 30, 60, 15, 80]`.
Use a `for` loop with `range` and `if` to print all items under ■50 and count how many are affordable.

Q3. AI Assignment Deadline Checker

A student submits an AI assignment. Ask for submission hour and minute. Deadline is 18:00.
If time is invalid, print `Invalid time`.
If submitted before or at 18:00, print `AI assignment submitted on time`.
Otherwise print `Late submission`.

Q4. Marks Calculator with Grace

Ask for marks of 3 subjects. Calculate average.
If any subject is below 30, print `Fail: one subject below 30`.
Else if average ≥ 90 print `Grade A`,
else if average ≥ 75 print `Grade B`,
else if average ≥ 50 print `Grade C`,
else print `Fail`.

Q5. Ethical Friend Phone Prank Simulator

You want to make a funny Python prank on a friend's phone, but only with permission.
Ask: `Do you have permission? yes/no`.
If no, print `Access denied. Ethical hacking needs consent.`
If yes, ask friend's name and battery percent.
If battery < 20, print a funny low-battery message. Else print a safe prank message.

Q6. Variable Type Detective

Create these variables: `token = 101`, `price = 45.5`, `food = "Poha"`, `is_veg = True`.
Print each value and its type. Then increase price by 10% and print the new value and type again.

Q7. Smart Q Delivery Speed

A Smart Q delivery person covers distances `[1.2, 0.8, 2.5]` km in times `[5, 4, 10]` minutes.
For each delivery, calculate speed in km/h. If speed > 15, print `Fast delivery`.

Q8. Swap Tokens

Create `a = "SmartQ"` and `b = "Cafeteria"`. Swap them without using a third variable and print.
Also use multiple assignment to set `x, y, z = 10, 20, 30` and print them.

Q9. Debug the Cafeteria Code

Find and correct the errors in this code:
`1food = "Samosa" price = 20 if price = 20 print("Cheap") else: print("Expensive")`

Q10. Free Dessert Condition

Write a condition for free dessert:
Free dessert if `(is_veg and price > 150) or (day == "Friday" and has_coupon)`.
Test with `is_veg = True, price = 180, day = "Monday", has_coupon = False`.
Print whether the student gets free dessert.

Q11. Range Token Queue

Use `range` to print token numbers from 101 to 110. Skip token 105.
Then print only even token numbers from the same range, also skipping 105.

Q12. Filter AI Assignment Submissions

Given `["submitted", "late", "submitted", "missing", "submitted", "late"]`, use a `for` loop and `if-elif-else` to count how many are submitted, late, and missing. Print the summary.

Q13. Cafeteria Bill with Tax and Discount

Ask for total amount.
If total > 500, give 10% discount; otherwise no discount.
Then apply 5% tax on the amount after discount.
Print total, discount, tax, and final bill.

Q14. Boolean Confusion

Given `A = True, B = False, C = True`, evaluate and print:
1. `not A and B or C`
2. `A or B and not C`
3. `(A and B) or (not B and C)`

Q15. Smart Q Order Validator

Menu is `["Poha", "Samosa", "Maggi", "Tea"]`.
Ask for item and quantity. If item is in the menu and quantity > 0, print `Order accepted`; else print `Invalid order`. Use a `for` loop to check the item, not direct `in`.

Q16. Ethical Hacking Password Checker

Ask for a friend's phone password in a simulated ethical hacking demo.
If password is `"1234"` or `"password"`, print `Weak: friend's phone could be hacked. Change it.`
Else if length is less than 8, print `Medium weak: use at least 8 characters.`
Else print `Strong password.`